

Nightlife.com.vn — Technical Specification

Version: 1.0 | **Date:** 2026 | **Stack:** Next.js 15 + Supabase + Redis + WebSocket

1. OVERVIEW

Nightlife.com.vn là nền tảng nightlife full-stack với 6 core features: 1. Venue discovery & booking (table/VIP room) 2. Events & ticket sales 3. Squad Night (group booking + bill splitting) 4. Happy Hour deals aggregator 5. Community forum (hỏi đáp đi bar) 6. VIP Membership (3 tiers)

2. DATABASE SCHEMA (Supabase / PostgreSQL)

2.1 Users & Membership

```
users (  
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  email       TEXT UNIQUE NOT NULL,  
  password_hash TEXT,  
  display_name TEXT NOT NULL,  
  full_name   TEXT,  
  avatar_url  TEXT,  
  phone       TEXT,  
  city        TEXT DEFAULT 'hcm',    -- 'hcm' | 'hanoi' | 'danang'  
  role        TEXT DEFAULT 'user',   -- 'user' | 'venue_owner' | 'admin'  
  membership_tier TEXT DEFAULT 'free', -- 'free' | 'night_pass' | 'black_card'  
  membership_expires_at TIMESTAMPTZ,  
  nightlife_passport_points INTEGER DEFAULT 0,  
  created_at  TIMESTAMPTZ DEFAULT NOW()  
)
```

2.2 Venues

```
venues (  
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  slug        TEXT UNIQUE NOT NULL,      -- 'observatory-saigon'  
  owner_id    UUID REFERENCES users(id),  
  name        TEXT NOT NULL,  
  type        TEXT NOT NULL,             -- 'rooftop_bar' | 'club' | 'wine_bar' | 'live_music'  
  description  TEXT,  
  address     TEXT NOT NULL,  
  district    TEXT,                     -- 'Q1' | 'Q2' | 'Q3'  
  city        TEXT DEFAULT 'hcm',  
  lat         DECIMAL(10,7),  
  lng         DECIMAL(10,7),  
  phone       TEXT,  
  website     TEXT,  
  instagram   TEXT,  
  cover_charge INTEGER DEFAULT 0,        -- VND, 0 = free entry  
  price_range TEXT DEFAULT '$$',        -- '$' | '$$' | '$$$' | '$$$$'  
  capacity    INTEGER,  
  min_spend   INTEGER,                  -- Minimum spend per table  
  dress_code  TEXT,  
  age_restriction INTEGER DEFAULT 18,  
  open_hours  JSONB,                    -- {"mon": "18:00-02:00", "fri": "18:00-04:00"}  
  features    TEXT[],                  -- ['live_dj', 'live_band', 'pool', 'outdoor']  
  thumbnail_url TEXT,  
  images      TEXT[],  
  is_verified BOOLEAN DEFAULT FALSE,  
  is_active   BOOLEAN DEFAULT TRUE,  
  is_vip_only BOOLEAN DEFAULT FALSE,  
  avg_rating  DECIMAL(3,2),  
  total_reviews INTEGER DEFAULT 0,  
  total_bookings INTEGER DEFAULT 0,  
  subscription_tier TEXT DEFAULT 'basic', -- 'basic' | 'premium' (B2B plan)  
  created_at  TIMESTAMPTZ DEFAULT NOW()  
)  
  
venue_tables (  
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  venue_id    UUID REFERENCES venues(id) ON DELETE CASCADE,  
  table_name  TEXT NOT NULL,           -- 'VIP Room 1' | 'Table A3'  
  type        TEXT DEFAULT 'standard', -- 'standard' | 'vip' | 'booth'  
  capacity    SMALLINT NOT NULL,
```

```

min_spend          INTEGER,
deposit_required   INTEGER DEFAULT 0,
is_active          BOOLEAN DEFAULT TRUE
)

```

2.3 Bookings

```

bookings (
  id              UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  venue_id        UUID REFERENCES venues(id),
  table_id        UUID REFERENCES venue_tables(id),
  user_id         UUID REFERENCES users(id),
  squad_id        UUID REFERENCES squads(id),      -- null nếu book cá nhân
  booking_date    DATE NOT NULL,
  booking_time    TIME NOT NULL,
  party_size      SMALLINT NOT NULL,
  status          TEXT DEFAULT 'pending',          -- 'pending'|'confirmed'|'seated'|'completed'
  special_requests TEXT,
  deposit_amount  INTEGER DEFAULT 0,
  deposit_paid    BOOLEAN DEFAULT FALSE,
  payment_ref     TEXT,
  confirmed_at    TIMESTAMPTZ,
  cancelled_at    TIMESTAMPTZ,
  created_at      TIMESTAMPTZ DEFAULT NOW()
)

```

2.4 Events & Tickets

```

events (
  id              UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  slug            TEXT UNIQUE NOT NULL,
  venue_id        UUID REFERENCES venues(id),
  title           TEXT NOT NULL,
  description      TEXT,
  event_date      DATE NOT NULL,
  start_time      TIME NOT NULL,
  end_time        TIME,
  genre           TEXT[],                      -- ['edm', 'hip_hop', 'jazz']
  lineup          TEXT[],                      -- ['DJ Tiesto', 'Local DJ']
  thumbnail_url   TEXT,
  images          TEXT[],
  is_free         BOOLEAN DEFAULT FALSE,
)

```

```

    age_restriction    INTEGER DEFAULT 18,
    total_capacity     INTEGER,
    is_active          BOOLEAN DEFAULT TRUE,
    created_at         TIMESTAMPTZ DEFAULT NOW()
)

ticket_tiers (
    id                 UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    event_id           UUID REFERENCES events(id) ON DELETE CASCADE,
    name               TEXT NOT NULL,          -- 'Early Bird' | 'General' | 'VIP'
    price              INTEGER NOT NULL,       -- VND
    quantity           INTEGER NOT NULL,       -- Số lượng vé loại này
    sold               INTEGER DEFAULT 0,
    includes           TEXT[],                -- ['1 welcome drink', 'priority entry']
    sale_starts_at     TIMESTAMPTZ,
    sale_ends_at       TIMESTAMPTZ
)

tickets (
    id                 UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tier_id             UUID REFERENCES ticket_tiers(id),
    event_id           UUID REFERENCES events(id),
    user_id            UUID REFERENCES users(id),
    order_id           UUID,
    ticket_code        TEXT UNIQUE NOT NULL,   -- QR code value
    status             TEXT DEFAULT 'valid',   -- 'valid' | 'used' | 'refunded'
    checked_in_at      TIMESTAMPTZ,
    created_at         TIMESTAMPTZ DEFAULT NOW()
)

ticket_orders (
    id                 UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id            UUID REFERENCES users(id),
    event_id           UUID REFERENCES events(id),
    total_amount       INTEGER NOT NULL,
    platform_fee       INTEGER NOT NULL,      -- 5% ticketing fee
    status             TEXT DEFAULT 'pending', -- 'pending' | 'paid' | 'refunded'
    payment_method     TEXT,
    payment_ref        TEXT,
    created_at         TIMESTAMPTZ DEFAULT NOW()
)

```

2.5 Squad Night

```
squads (  
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  invite_code TEXT UNIQUE NOT NULL,    -- 6-char code để share  
  created_by  UUID REFERENCES users(id),  
  venue_id    UUID REFERENCES venues(id),  
  booking_date DATE,  
  booking_time TIME,  
  party_size  SMALLINT,  
  budget_per_person INTEGER,  
  status      TEXT DEFAULT 'forming', -- 'forming'|'confirmed'|'completed'  
  bill_splitting BOOLEAN DEFAULT TRUE,  
  total_bill  INTEGER,  
  created_at  TIMESTAMPTZ DEFAULT NOW()  
)  
  
squad_members (  
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  squad_id     UUID REFERENCES squads(id) ON DELETE CASCADE,  
  user_id      UUID REFERENCES users(id),  
  role         TEXT DEFAULT 'member', -- 'host' | 'member'  
  status       TEXT DEFAULT 'invited', -- 'invited'|'confirmed'|'declined'  
  share_amount INTEGER,                -- phần thanh toán của người này  
  paid         BOOLEAN DEFAULT FALSE,  
  joined_at    TIMESTAMPTZ DEFAULT NOW()  
)
```

2.6 Happy Hour Deals

```
deals (  
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  venue_id     UUID REFERENCES venues(id) ON DELETE CASCADE,  
  title        TEXT NOT NULL,          -- 'House Cocktail -50%'  
  description  TEXT,  
  discount_type TEXT,                  -- 'percent' | 'fixed' | 'b1g1' | 'free_entry'  
  discount_value INTEGER,              -- 50 nếu percent, 50000 nếu fixed  
  applicable_days TEXT[],              -- ['mon','tue','wed','thu','fri','sat','sun']  
  start_time   TIME NOT NULL,  
  end_time     TIME NOT NULL,  
  conditions   TEXT,                  -- 'Áp dụng khi đặt bàn trước qua Nightlife.vn'  
  is_exclusive BOOLEAN DEFAULT FALSE, -- chỉ dành cho VIP member
```

```

    is_active          BOOLEAN DEFAULT TRUE,
    valid_until         DATE,
    created_at          TIMESTAMPTZ DEFAULT NOW()
)

```

2.7 Community Forum

```

forum_posts (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id           UUID REFERENCES users(id),
    title             TEXT NOT NULL,
    content           TEXT NOT NULL,
    city             TEXT,
    tags             TEXT[],                -- ['rooftop', 'Q1', 'group', 'review']
    venue_id          UUID REFERENCES venues(id), -- nếu review cụ thể 1 venue
    is_pinned         BOOLEAN DEFAULT FALSE,
    is_approved       BOOLEAN DEFAULT TRUE,
    view_count        INTEGER DEFAULT 0,
    reply_count       INTEGER DEFAULT 0,
    created_at        TIMESTAMPTZ DEFAULT NOW()
)

forum_replies (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    post_id           UUID REFERENCES forum_posts(id) ON DELETE CASCADE,
    user_id           UUID REFERENCES users(id),
    parent_id         UUID REFERENCES forum_replies(id), -- nested reply
    content           TEXT NOT NULL,
    is_approved       BOOLEAN DEFAULT TRUE,
    helpful_count     INTEGER DEFAULT 0,
    created_at        TIMESTAMPTZ DEFAULT NOW()
)

```

2.8 Bar Tour (ĐNA Packages)

```

tour_packages (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    slug              TEXT UNIQUE NOT NULL,
    country           TEXT NOT NULL,        -- 'thailand' | 'indonesia' | 'singapore'
    city              TEXT NOT NULL,        -- 'Bangkok' | 'Bali' | 'Singapore'
    title             TEXT NOT NULL,
    description       TEXT,

```

```

    duration_days    SMALLINT NOT NULL,
    price_per_person INTEGER NOT NULL,
    includes         TEXT[],          -- ['khách sạn 4*', '3 venues']
    highlights       TEXT[],          -- Tên các venues nổi bật
    thumbnail_url    TEXT,
    images           TEXT[],
    max_group_size   SMALLINT DEFAULT 10,
    is_active        BOOLEAN DEFAULT TRUE,
    created_at       TIMESTAMPTZ DEFAULT NOW()
)

tour_bookings (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    package_id        UUID REFERENCES tour_packages(id),
    user_id           UUID REFERENCES users(id),
    travel_date       DATE,
    group_size        SMALLINT DEFAULT 1,
    total_amount      INTEGER,
    status            TEXT DEFAULT 'inquiry', -- 'inquiry'|'confirmed'|'cancelled'
    notes             TEXT,
    created_at        TIMESTAMPTZ DEFAULT NOW()
)

```

2.9 Venue Reviews

```

venue_reviews (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    venue_id          UUID REFERENCES venues(id) ON DELETE CASCADE,
    user_id           UUID REFERENCES users(id),
    booking_id        UUID REFERENCES bookings(id), -- verify đã đến
    rating            SMALLINT NOT NULL CHECK (rating BETWEEN 1 AND 5),
    atmosphere_rating SMALLINT,
    service_rating    SMALLINT,
    value_rating      SMALLINT,
    content           TEXT,
    visited_date      DATE,
    images            TEXT[],
    is_verified_visit BOOLEAN DEFAULT FALSE,
    helpful_count     INTEGER DEFAULT 0,
    created_at        TIMESTAMPTZ DEFAULT NOW()
)

```

3. API ENDPOINTS

Base URL: `/api/v1`

3.1 Auth

```
POST    /auth/register
POST    /auth/login
POST    /auth/logout
POST    /auth/refresh
GET     /auth/me
PUT     /auth/profile      -- Cập nhật profile + avatar
```

3.2 Venues

```
GET     /venues            -- Danh sách venues
Query params:
  city=hcm
  type=rooftop_bar
  district=Q1
  price_range=$$
  features=live_dj
  is_open_now=true          -- Filter đang mở cửa
  sort=rating|popular|newest
  page=1&limit=20

GET     /venues/:slug      -- Chi tiết venue + tables + deals + reviews
GET     /venues/:slug/availability
Query:  date=2026-05-24&party_size=4
Response: available tables + time slots

GET     /venues/:slug/reviews -- Reviews của venue
POST    /venues/:slug/reviews -- Tạo review (cần auth + verified booking)
GET     /venues/nearby     -- Venues gần vị trí hiện tại
Query:  lat=10.77&lng=106.69&radius=2000
```

3.3 Bookings

```
POST    /bookings          -- Tạo booking
Body: {
```



```
venue_id, table_id, booking_date, booking_time,  
party_size, special_requests, payment_method  
}
```

```
GET    /bookings/mine          -- Bookings của tôi  
GET    /bookings/:id          -- Chi tiết booking  
PUT    /bookings/:id/cancel -- Hủy booking (nếu > 2h trước)  
POST   /bookings/:id/checkin  -- Venue check-in (venue owner)
```

3.4 Events & Tickets

```
GET    /events              -- Danh sách events  
      Query: city=hcm&date_from=2026-05-01&genre=edm  
  
GET    /events/:slug       -- Chi tiết event + ticket tiers  
POST   /events/:slug/purchase -- Mua vé  
      Body: { tier_id, quantity, payment_method }  
  
GET    /tickets/mine       -- Vé của tôi  
GET    /tickets/:code      -- Xem vé (dùng cho QR scan)  
POST   /tickets/:code/checkin -- Venue scan QR check-in
```

3.5 Squad Night

```
POST   /squads              -- Tạo squad mới  
      Body: { venue_id, booking_date, booking_time, party_size, budget_per_person }  
      Response: { squad_id, invite_code, invite_url }  
  
GET    /squads/:invite_code -- Xem thông tin squad (public để share)  
POST   /squads/:invite_code/join -- Tham gia squad  
PUT    /squads/:id/confirm  -- Host confirm → tạo booking  
POST   /squads/:id/split-bill -- Tính toán chia bill  
      Body: { total_amount }  
      Response: { per_person_amount, payment_links }  
  
POST   /squads/:id/pay      -- Từng member thanh toán phần của mình  
GET    /squads/mine         -- Squads của tôi (host + member)
```

3.6 Deals

```
GET    /deals          -- Danh sách Happy Hour deals active hôm nay
      Query: city=hcm&district=Q1&is_open_now=true

GET    /deals/venue/:venue_id -- Deals của 1 venue cụ thể
```

3.7 Forum

```
GET    /forum/posts      -- Danh sách posts
      Query: city=hcm&tag=rooftop&sort=hot|new|top

POST   /forum/posts      -- Tạo post mới (cần auth)
GET    /forum/posts/:id  -- Chi tiết post + replies
POST   /forum/posts/:id/replies -- Trả lời
POST   /forum/posts/:id/report -- Báo cáo
```

3.8 Membership

```
GET    /membership/tiers -- Thông tin 3 gói VIP
POST   /membership/subscribe -- Đăng ký VIP
      Body: { tier: 'night_pass' | 'black_card', payment_method }

GET    /membership/mine -- Thông tin gói hiện tại
POST   /membership/cancel -- Hủy auto-renewal
```

3.9 Passport & Gamification

```
POST   /passport/checkin -- Check-in tại venue (tích điểm)
      Body: { venue_id, booking_id }

GET    /passport/mine -- Điểm + lịch sử check-in
GET    /passport/rewards -- Danh sách rewards có thể đổi
POST   /passport/redeem -- Đổi điểm lấy rewards
```

3.10 Bar Tours

```
GET    /tours                -- Danh sách tour packages
GET    /tours/:slug         -- Chi tiết tour
POST   /tours/:slug/inquiry -- Gửi yêu cầu đặt tour
```

3.11 Payments & Webhooks

```
POST   /payments/vnpay/create
POST   /payments/vnpay/ipn      -- WEBHOOK (không cần auth)
POST   /payments/momo/create
POST   /payments/momo/ipn      -- WEBHOOK (không cần auth)
POST   /payments/stripe/create -- Cho khách nước ngoài
POST   /payments/stripe/webhook -- WEBHOOK
```

4. BUSINESS LOGIC

4.1 Booking Flow

1. User chọn venue + ngày giờ + số người
2. API check availability → trả về available tables
3. User chọn bàn + điền thông tin
4. Nếu có deposit → redirect payment → webhook confirm
5. Booking status: pending → confirmed (sau payment / sau venue confirm)
6. Venue nhận notification (email + push) để chuẩn bị
7. User nhận confirmation email + calendar invite
8. Ngày đi: venue check-in → passport points +10
9. 24h sau → reminder email để review

4.2 Squad Night Flow

1. Host tạo squad → nhận invite_code (6 ký tự)
2. Host share link: nightlife.vn/squad/ABC123
3. Bạn bè click link → đăng nhập → join squad
4. Khi đủ người (hoặc host quyết định) → host confirm
5. System tạo booking tự động với party_size = số member
6. Chia bill:
 - Host nhập total_bill sau khi đi về

- System tính `per_person = total / members`
 - Mỗi người nhận payment link (VNPay/MoMo)
 - Mark paid khi webhook confirm
7. Tích điểm Passport cho cả squad

4.3 Ticket Sales Flow

1. User chọn tier + quantity
2. Check inventory: `tier.quantity - tier.sold >= quantity`
3. Tạo order → lock seats (5 phút để thanh toán)
4. Redirect payment
5. Webhook confirm → generate ticket codes (UUID-based QR)
6. Email vé PDF với QR code
7. Ngày event: venue scan QR → API validate → mark `checked_in`

4.4 VIP Membership Perks

Night Pass (199k/tháng):

- Priority booking: thấy availability 48h trước free users
- Exclusive deals: `access deals is_exclusive = TRUE`
- Free cover charge: 2 vouchers/tháng (cộng vào membership)
- Squad discount: -20% khi `party_size >= 4`

Black Card (499k/tháng):

- Tất cả Night Pass perks
- Guaranteed VIP table: nếu không còn thì venue phải offer alternative
- Concierge hotline: số điện thoại riêng để book 24/7
- Unlimited free cover charge
- Bar Tour discount: -15% tất cả packages

4.5 Passport Points System

Earn points:

- Check-in tại venue: +10 points
- Review sau khi đến: +5 points
- Đặt vé event: +15 points
- Squad Night (cả nhóm): +20 points host, +10 points members
- Giới thiệu bạn đăng ký: +50 points

Redeem rewards:

- 100 points → 1 free drink voucher

- 300 points → free cover charge 1 venue
- 500 points → 1 tháng Night Pass
- 1000 points → 1 tháng Black Card

5. REALTIME FEATURES (Supabase Realtime)

```
// 1. Live venue availability – cập nhật khi có booking mới
supabase
  .channel('venue-availability')
  .on('postgres_changes', {
    event: 'INSERT',
    schema: 'public',
    table: 'bookings',
    filter: `venue_id=eq.${venueId}`
  }, payload => {
    // Update available tables count on UI
  })

// 2. Squad real-time updates – member join/leave
supabase
  .channel(`squad-${squadId}`)
  .on('postgres_changes', {
    event: '*',
    schema: 'public',
    table: 'squad_members',
    filter: `squad_id=eq.${squadId}`
  }, payload => {
    // Update member list + bill calculation
  })

// 3. Live event checkin count (cho venue staff)
supabase
  .channel(`event-${eventId}-checkins`)
  .on('postgres_changes', {
    event: 'UPDATE',
    schema: 'public',
    table: 'tickets',
    filter: `event_id=eq.${eventId}`
  }, payload => {
```

```
// Update checkin counter for venue staff app
})
```

6. AUTH & SECURITY

Authentication: JWT (access 15 phút + refresh 30 ngày)
Social login: Google OAuth (nhiều user nightlife dùng Google)

Rate limits:

- Booking API: 10 req/min per user (tránh spam đặt bàn)
- Forum post: 5 posts/hour per user
- Public API: 100 req/min per IP

Row Level Security:

- bookings: chỉ đọc/sửa của mình hoặc venue_owner
- squad_members: chỉ đọc squad mình tham gia
- forum_posts: read all (approved), write của mình
- venue tables: read all, write chỉ venue_owner

Venue owner permissions:

- Quản lý tables, deals, events của venue mình
- Xem bookings của venue mình
- Check-in bookings và scan QR vé
- KHÔNG được đọc personal data của user

7. THIRD-PARTY INTEGRATIONS

Service	Mục đích	Ghi chú
Supabase	DB + Auth + Storage + Realtime	Primary
Redis (Upstash)	Cache venue list, deals, availability	TTL 5 phút cho availability
VNPay	Thanh toán booking + vé + squad	Phổ biến nhất VN

Service	Mục đích	Ghi chú
MoMo	Ví điện tử	Tiện cho thanh toán nhanh
Stripe	Khách nước ngoài (expat)	Card payment
Cloudinary	Image CDN	Venue photos
Resend	Email confirm + vé PDF	
Firebase Cloud Messaging	Push notification	Reminder, squad updates
Google Maps API	Venue map, nearby search	
BookingModel API	DJ/Talent booking từ venue	Internal VEA API
VEA Retail API	Ticket sales channel	Internal VEA API

8. SEO REQUIREMENTS

```

/ → Trang chủ
/venue/[slug] → LocalBusiness schema + Review schema
/event/[slug] → Event schema (schema.org/Event)
/bai-viet/[slug] → Blog posts cho SEO
/[city]/bar → "bar Q1 hcm" local SEO
/[city]/club → "club saigon" local SEO
/[city]/happy-hour → "happy hour saigon" keyword

```

Dynamic sitemap:

- /venue/* (2800+ URLs)
- /event/* (120+ mỗi tháng)
- /bai-viet/*

Google My Business integration:

- Sync venue data với Google Business API
- Import reviews 2 chiều

9. NOTIFICATIONS

Booking confirmed	→ Email + Push
Booking reminder	→ Push (3h trước)
Squad member joined	→ Push realtime
Squad bill ready	→ Push + Email với payment link
Event starts tomorrow	→ Push
Happy Hour starting	→ Push (15 phút trước, opt-in)
New forum reply	→ Push
Passport milestone	→ Push (100, 300, 500, 1000 points)
Membership expiring	→ Email (7 ngày trước)

10. INFRASTRUCTURE

Frontend:	Vercel (Next.js) – Vietnam Edge Network
Database:	Supabase Pro – Singapore region
Cache:	Upstash Redis – Singapore region
Images:	Cloudinary
Push:	Firebase Cloud Messaging (FCM)
Email:	Resend
Maps:	Google Maps Platform

Environment variables:

```
NEXT_PUBLIC_SUPABASE_URL
NEXT_PUBLIC_SUPABASE_ANON_KEY
SUPABASE_SERVICE_ROLE_KEY
REDIS_URL
VNPAY_TMN_CODE
VNPAY_HASH_SECRET
MOMO_PARTNER_CODE
MOMO_ACCESS_KEY
MOMO_SECRET_KEY
STRIPE_SECRET_KEY
STRIPE_WEBHOOK_SECRET
CLOUDINARY_URL
RESEND_API_KEY
FIREBASE_SERVICE_ACCOUNT
```


GOOGLE_MAPS_API_KEY
NEXT_PUBLIC_SITE_URL

11. ESTIMATED TIMELINE

Phase	Tasks	Estimate
1	Setup + schema migration + seed data	2 ngày
2	Auth + user profile + Google OAuth	2 ngày
3	Venue CRUD + search + filter + maps	4 ngày
4	Booking system + availability check	3 ngày
5	Payment integration (VNPay + MoMo)	3 ngày
6	Events + ticket sales + QR checkin	4 ngày
7	Squad Night + bill splitting	3 ngày
8	Happy Hour deals system	1 ngày
9	Forum community	2 ngày
10	VIP Membership + Passport points	3 ngày
11	Realtime (Supabase Realtime)	2 ngày
12	Push notifications (FCM)	2 ngày
13	Bar Tour module	2 ngày
14	Venue owner dashboard	3 ngày
15	SEO + sitemap + schema markup	2 ngày
16	Testing + QA + deploy	3 ngày
Total		~41 ngày